Melanie Fernández

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Education

Connecticut College New London, CT
Bachelor of Arts in Computer Science, Minor in Sociology *Aug. 2022 – May 2026*

- **Focus:** Machine Learning and Artificial Intelligence
- **Relevant Courses:** Data Structures, Computational Intelligence, Computer Organization, Algorithms, Data Visualization, Entertainment Software Design, Database Systems, Computer Networks.

Technical Skills

Languages: Python, Java, SQL, JavaScript, R, C#, C++, HTML
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, reinforcement learning, meta-learning, neural networks
Data & Databases: MySQL, SQLite, pandas, NumPy, structured dataset management, data cleaning, data validation
Systems & Development: Linux/Bash, Git, FastAPI, Flask, Django, RESTful APIs, Flutter

Publications

O'Connor, J. and Fernández, M. (2024). *Porto, Portugal.*
Online Match Prediction in Shogi Using Deep Convolutional Neural Networks. Proceedings of the International Conference on Neural Computation Theory and Applications (IJCCI). **Nominated Best Student Paper.**

O'Connor, J. and Fernández, M. (2025).
Advancements in Online Match Prediction in Shogi Using Deep Convolutional Neural Networks.
Post-publication extension, Computational Intelligence, Springer.

Parker, G., O'Connor, J., Nash, J., and Fernández, M. (2026). *Marbella, Spain.*
Simulation Speciation and Complex Trait Evolution in Multi-Species Artificial Environment. Proceedings of the International Conference on Agents and Artificial Intelligence.

O'Connor, J. and Fernández, M. (2026). *Pending Publication, SMC 2026.*
Opening-as-Task Meta-Learning for Cross-Game Position Evaluation in Chess and Shogi. Built a cross-game meta-learning model using ANIL to learn transferable representations between Chess and Shogi, enabling stable convergence with minimal task-specific parameter updates.

O'Connor, J. and Fernández, M. (2026). *Pending Publication, CIFEr 2026.*
MAGE: MAP-Elites for Alpha Generation. Applied MAP-Elites and genetic programming to generate diverse and robust alpha trading factors on S&P 500 market data for quantitative finance research.

Research & Technical Experience

Independent Software Developer May 2026 – Present
Contract Project for Medical Study Prototype *Remote*

- Designing and developing a tablet-based survey platform to support behavioral and psychological research studies.
- Building a cross-platform Flutter application with structured survey workflows and spreadsheet-compatible data export.
- Collaborating directly with researchers to translate study requirements into deployable software tools.
- Implementing response-handling workflows to improve data consistency, organization, and future analysis.

Undergraduate Research Assistant June 2023 – May 2026
Autonomous Agent Learning Lab, Connecticut College *New London, CT*

- Conduct research in deep learning, reinforcement learning, and meta-learning under faculty supervision.
- Designed and implemented convolutional neural network architectures for board-game position evaluation and match prediction.
- Built Python-based data pipelines to preprocess and transform large-scale game datasets for machine learning experiments.
- Developed training and evaluation workflows using PyTorch, NumPy, pandas, and scikit-learn.

- Co-authored peer-reviewed publications on deep learning applications in Shogi and strategic game environments.
- Investigating Model-Agnostic Meta-Learning and ANIL-based approaches for knowledge transfer between Chess and Shogi.
- Collaborated with researchers to improve experimental reproducibility, document results, and communicate findings.

Research Fellow August 2024 – December 2024
SZTAKI MILAB, AIT Budapest *Budapest, Hungary*
- Built end-to-end Python pipelines for dataset cleaning, validation, and transformation at scale.
- Implemented KNN and regression models to recover missing data values, improving dataset completeness from 62% to 95%.
- Evaluated model reliability and collaborated with researchers to analyze computational patterns in structured datasets.

Data Analyst September 2022 – December 2022
Connecticut College *New London, CT*
- Cleaned, structured, and validated alumni datasets using SQL and Excel to support institutional reporting and analysis.
- Standardized data categorization procedures and documentation practices to improve long-term traceability and dataset reuse.
- Performed data quality checks and validation workflows to ensure consistency across institutional records.

Teaching Experience

Teaching Assistant Sept. 2024 – May 2026
Connecticut College *New London, CT*
- Supported undergraduate students in Computational Intelligence, Algorithms, and computer science coursework.
- Provided one-on-one guidance on algorithmic problem solving, machine learning concepts, and programming assignments.
- Assisted students in debugging code, understanding technical concepts, and developing stronger problem-solving strategies.

Projects

SupportHub — Role-Based Ticketing System (Full-Stack) *GitHub: HR_API*
- Built a full-stack ticketing system using Django REST Framework with JWT authentication and role-based access control for customers, staff, and managers.
- Designed RESTful API architecture and a JavaScript/HTML frontend; wrote structured data models and query abstractions to support scalable backend operations.

Interactive Data Visualization for Star Data *GitHub: Star Visualization*
- Developed interactive visualizations using JavaScript and D3.js to explore large astronomical datasets with dynamic filtering and real-time user-driven analysis.

Honors and Awards

Nomination, Best Student Paper, IJCCI *Porto, Portugal*
International Conference on Neural Computation Theory and Applications.

Science Leaders Program *Connecticut College*
Selected as a member of an honors cohort promoting the integration of scientific and social research.